EXPERIMENT 11

Simple Top-Down Parser for Context-Free Grammars (CFG)

AIM

To implement a recursive descent top-down parser for Context-Free Grammars in Python to validate and parse grammatical sentences.

PROCEDURE

1. Define Context-Free Grammar production rules starting from root non-terminal S.
2. Represent grammar rules using NLTK's `CFG.fromstring()`.
3. Initialize the `RecursiveDescentParser`.
4. Supply candidate tokenized strings to the parser.
5. Traverse top-down from S matching productions to input tokens and output parsed trees.

PROGRAM

```
import nltk
from nltk import CFG

cfg_grammar = CFG.fromstring("""
    S -> NP VP
    VP -> V NP | V
    NP -> Det N | N
    Det -> 'the' | 'a'
    N -> 'cat' | 'dog' | 'telescope' | 'astronomer'
    V -> 'saw' | 'chased'
""")

parser = nltk.RecursiveDescentParser(cfg_grammar)

test_sentences = [
    "the cat chased a dog",
    "the astronomer saw the cat"
]

print("--- Top-Down Parsing Trace ---")
for sent in test_sentences:
    tokens = sent.split()
    print(f"\nParsing Sentence: '{sent}'")
    parsed_trees = list(parser.parse(tokens))
    if parsed_trees:
        print(f"Success! {len(parsed_trees)} parse tree(s) found:")
        for tree in parsed_trees:
            print(tree)
    else:
        print("Parsing Failed: Sentence does not conform to grammar.")
```

OUTPUT

```
> python3 experiment_11.py
--- Top-Down Parsing Trace ---

Parsing Sentence: 'the cat chased a dog'
Success! 1 parse tree(s) found:
(S (NP (Det the) (N cat)) (VP (V chased) (NP (Det a) (N dog))))

Parsing Sentence: 'the astronomer saw the cat'
Success! 1 parse tree(s) found:
(S (NP (Det the) (N astronomer)) (VP (V saw) (NP (Det the) (N cat))))
[Process completed with exit code 0]
```

RESULT

Thus, the top-down recursive descent parser for Context-Free Grammars was successfully implemented and validated.